جامعة الزرقاء
ZARQA UNIVERSITY

Department of Computer Science

Course Name & No.: Cloud Computing 1501438

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Ch 3- Benefits and Challenges

Assignment 3

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Task 1: Benefits of Cloud Computing
List five benefits of cloud computing as mentioned in the chapter and provide a one-sentence explanation for each.

1. Benefit:
   Explanation:

2. Benefit:
   Explanation:

3. Benefit:
   Explanation:

4. Benefit:
   Explanation:

5. Benefit:
   Explanation:

Task 2: Challenges in Cloud Computing
Identify three challenges mentioned in the chapter and briefly describe why each is a concern.

1. Challenge:
   Reason:

2. Challenge:
   Reason:

3. Challenge:
   Reason:

Task 3: Role of Web Services

Explain in your own words why web services are important in cloud computing. (2–3 sentences.)

Task 4: SOAP vs. REST

Compare SOAP-based web services and REST-compliant web services by filling in the table below:

Feature	SOAP-based Web Services	REST-compliant Web Services
Message Format		
Complexity		
Communication Method		
Primary Use Case		

Task 5: Role of APIs in Cloud

What is an API and how does it support cloud services?
Provide two examples of how APIs are used in cloud computing.

Definition:
Example 1:
Example 2:

Task1

 Benefit: Cost Efficiency
Explanation: Cloud computing reduces the need for expensive hardware and maintenance by using pay-as-you-go services.

 Benefit: Scalability
Explanation: Resources can be easily increased or decreased based on demand without physical upgrades.

 Benefit: Accessibility
Explanation: Users can access applications and data from anywhere with an internet connection.

 Benefit: Reliability
Explanation: Cloud providers offer backup and disaster recovery to ensure data is safe and available.

 Benefit: Automatic Updates
Explanation: Systems and software are automatically updated by the provider, saving time and effort.

Task2

 Challenge: Security and Privacy
Reason: Storing sensitive data on remote servers increases the risk of data breaches and unauthorized access.

 Challenge: Downtime
Reason: Cloud services depend on internet connectivity and provider stability, so outages can disrupt access.

 Challenge: Limited Control
Reason: Users have less control over infrastructure and data management compared to on-premise systems.

Task3

Web services are important in cloud computing because they allow different applications and systems to communicate over the internet. They enable data exchange and integration between services, making cloud platforms flexible and scalable. Without web services, connecting different cloud applications would be much more difficult.

Task4

Feature	**SOAP-based Web Services**	**REST-compliant Web Services**
Message Format	XML only	JSON, XML, or others
Complexity	More complex and strict	Simpler and lightweight
Communication Method	Uses protocols like HTTP, SMTP	Mainly uses HTTP methods (GET, POST, PUT, DELETE)
Primary Use Case	Enterprise-level, secure transactions	Web and mobile applications

Task5

Definition:
An API (Application Programming Interface) is a set of rules that allows different software applications to communicate and interact with each other.

Example 1:
A cloud storage API (like Google Drive API) allows applications to upload, download, and manage files in the cloud.

Example 2:
A cloud computing API (like AWS EC2 API) allows developers to create and manage virtual servers remotely.